

DEFECTS IN CRYSTALS

- ❑ Point defects ➤ 0D
- ❑ Line defects ➤ 1D
- ❑ Surface Imperfections ➤ 2D
- ❑ Volume Defects ➤ 3D

Part of

MATERIALS SCIENCE
& *A Learner's Guide*
ENGINEERING

AN INTRODUCTORY E-BOOK

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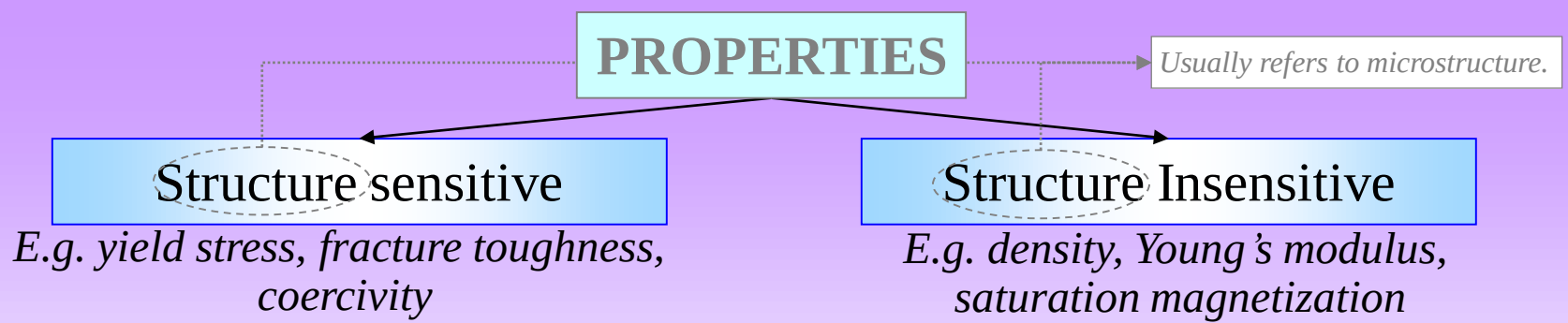
<http://home.iitk.ac.in/~anandh/E-book.htm>

Crystal Defects and Crystalline Interfaces

W. Bollmann

Springer-Verlag, New York (1970)

Caution Note: In any chapter, amongst the first few pages (say 5 pages) there will be some 'big picture' overview information. This may lead to 'overloading' and readers who find this 'uncomfortable' may skip particular slides in the first reading and come back to them later.



- ❑ Properties are classified into *Structure Sensitive* and *Structure Insensitive* properties.
- ❑ The key word to note is *sensitive* and **not** *dependent*.
 - E.g. density would be *dependent* on the concentration of vacancies. But, usually the concentration of vacancies is small and density would not be *sensitive* to the presence of vacancies.
 - Another example would be. Young's modulus (Y). ' Y ' would not be a sensitive function of the dislocation density
 - On the other hand, a structure sensitive property like yield stress, would be strongly dependent on the presence (or absence of dislocations). The yield stress in the absence of dislocations (in metals) would be typically of the order of GPa and in the presence of dislocations it would become of the order of MPa (*reduction by a few orders of magnitude*)!
- ❑ In the usual sense the word STRUCTURE means **MICROSTRUCTURE** (*and not crystal structure, nuclear structure, etc.*).
- ❑ In case of structure sensitive properties, the *Defect Structure* in the material plays an important role in determining the properties.

Importance of defects in crystals (*some examples*)

We will learn the details later

- ❑ In the absence of dislocations the yield strength of metals would be of the order of GPa. Dislocations severely weaken crystals and the yield strength falls to the order of MPa.
- ❑ In the absence vacancies, substitutional diffusion practically does not occur. In the presence of vacancies, this becomes feasible. If vacancies in excess of the equilibrium concentration are quenched-in, then the diffusivity increases by orders of magnitude.
- ❑ The growth of certain crystal facets from a liquid solution can be 'prohibitively' slow. The presence of surface steps created by terminating screw dislocations, provides sites for atomic attachment and the growth rate can increase by a 'humungous' factor.
- ❑ Colourless crystals develop colours in the presence of F-centres. E.g. violet colour in CaF_2 . Colour arises due to anionic vacancy in a crystal lattice being occupied electron(s), which leads to an absorption in the visible region.
- ❑ In TWIP (twinning induced plasticity) steels, the twinning process in conjunction with dislocation mediated plasticity, gives rise to a material with a combination of high strength and ductility.
- ❑ ➤ If thermal vibration of atoms (i.e. deviation from lattice sites) is considered a defect, then this effect is the major contributor to thermal expansion of metals.
➤ Metals like Cu are good conductors of both heat and electricity. The motion of electrons mediate the conduction of both heat and electricity. Diamond is a poor conductor of electricity, but a good conductor of heat. This is facilitated by atomic vibrations (**phonons**-collective oscillations of atoms).

What is meant by Defect Structure?

- ❑ The term *Defect Structure* hides in it a lot of details (similar to the word *Microstructure*) and a lot of parameters have to be specified to characterize this term (*and then try and understand its effect on the properties*).
- ❑ The following points go on to outline 'Defect Structure'.
 - Kinds of defects present along with their **dimensionality** (vacancies, dislocations, grain boundaries etc.).
 - The nature of these defects in terms of their origin: **Statistical** or **Structural**.
 - The nature of these defects in terms of their position: **Random** or **Ordered**.
 - **Density** and **spatial distribution** of these defects.
 - Interaction and association of these defects with each other.

Needless to say the task of understanding properties based on the defect structure is very difficult. The starting point would be to look at each defect in isolation and then put together parts of the picture.

Concept of **Defect in a Defect** & **Hierarchy of Defects**

[Click here to know more about Association of Defects](#)

[Click here to know more about Defect in a Defect](#)

Path to understanding Defect Structure

Take an isolated defect

Stress fields, charges, energy etc.

Consider pair-wise interaction of defects

Short range interactions (Stress fields, energy, charge)*

Behaviour of the entire 'defect structure' with external constraints

*Long range interactions & collective behaviour & external constraints***

- *Examples of pair-wise interactions would include:
 - Vacancy-vacancy interaction leading to the formation of a di-vacancy
 - Vacancy cluster's interaction with an vacancy leading to a larger vacancy cluster
 - Dislocation interstitial solute interaction leading to the formation of a “Cottrell atmosphere”
- **This is a difficult problem of materials science
 - Example would include the collective motion of dislocations (along with their interactions) leading to plastic deformation and work hardening

How can we classify defects in materials?

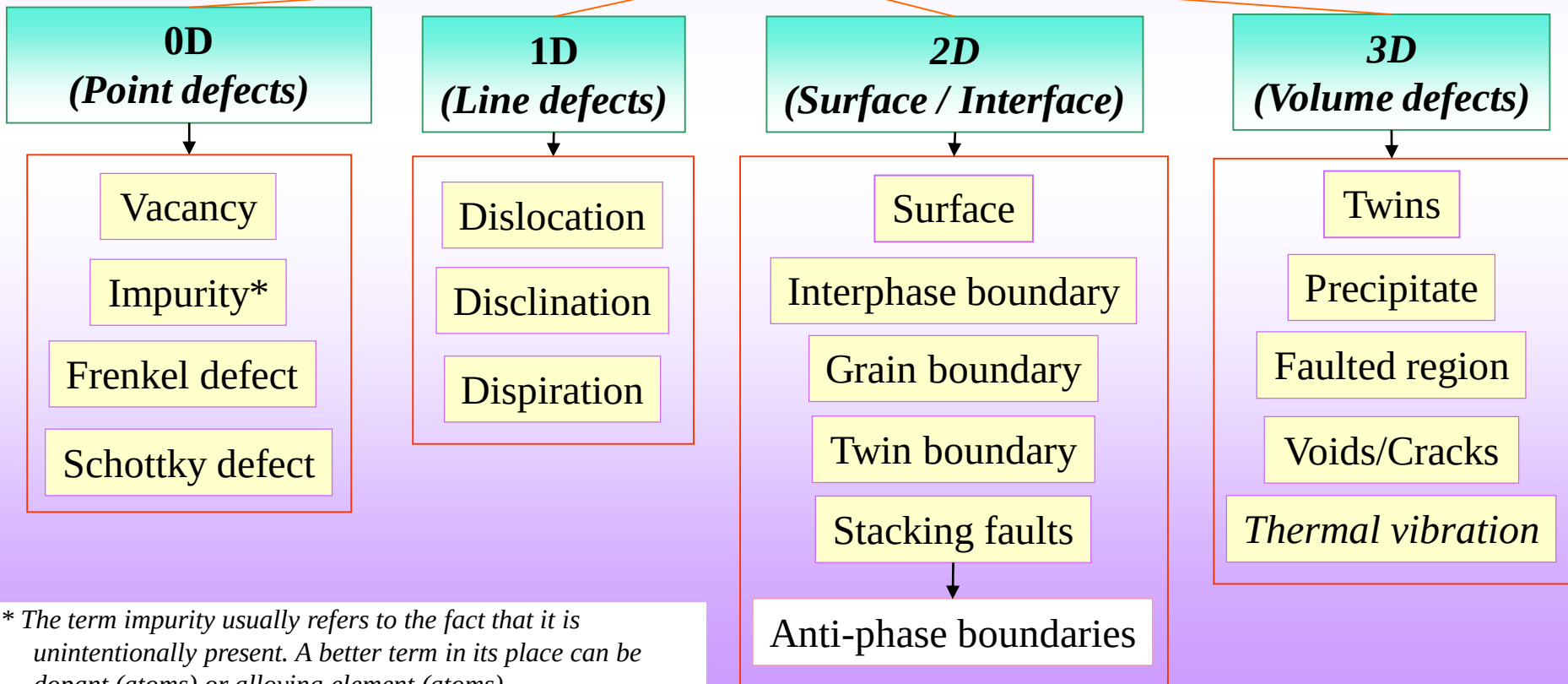
- ❑ Defects can be classified based on some of the following methods:
- Dimensionality
- Based on association with Symmetry and Symmetry Breaking
- Based on their origin
- Based on their position
- Based on the fact that if the defect is with respect to a geometrical entity or a physical property

In an elementary text it may not be practical to consider all the possibilities in detail. But, the student should keep in mind the possibilities and some of their implications on the properties or phenomena.

Classification Based on Dimensionality

- ❑ Truly speaking any defect exists in 3D. However, the ‘effective dimension’ may be lower. E.g. the strain field of a dislocation is in 3D, but it is a ‘line-like’ defect. Similarly, a vacancy is point-like.
- ❑ In special circumstances the dimension of defect may be lowered (e.g. in a 2D crystal a dislocation is point or a crack may be planar (2D)).

CLASSIFICATION OF DEFECTS BASED ON DIMENSIONALITY



Classification of defects based on their association with symmetry

- ❑ Clearly a defect will 'break' the perfect symmetry of a crystal. However, if the concentration of these defects is small, we assume that the crystal is perfect elsewhere, except in the vicinity of the defect (i.e. we continue to treat the structure as a crystal).
- ❑ At the atomic level, we can associate defects with translational, rotational and screw symmetries as in the figure below. At a larger scale, we can have domains in the crystal related to other domains across an interface via symmetry operators like: mirror, rotation or inversion (figure below).

SYMMETRY ASSOCIATED DEFECTS

Translation

Rotation

Screw

Dislocation

Disclination

Dispiration

SYMMETRY ASSOCIATED DEFECTS

Mirror

Rotation

Inversion

Twins

The Operation Defining a Defect Cannot be a Symmetry Operation of the Crystal

E.g. a twin plane in a mirror twin cannot be a mirror plane of the crystal

Atomic Level

Multi-atom

*Hence association
with symmetry*

**Based on
Symmetry
breaking**

DEFECTS

Topological

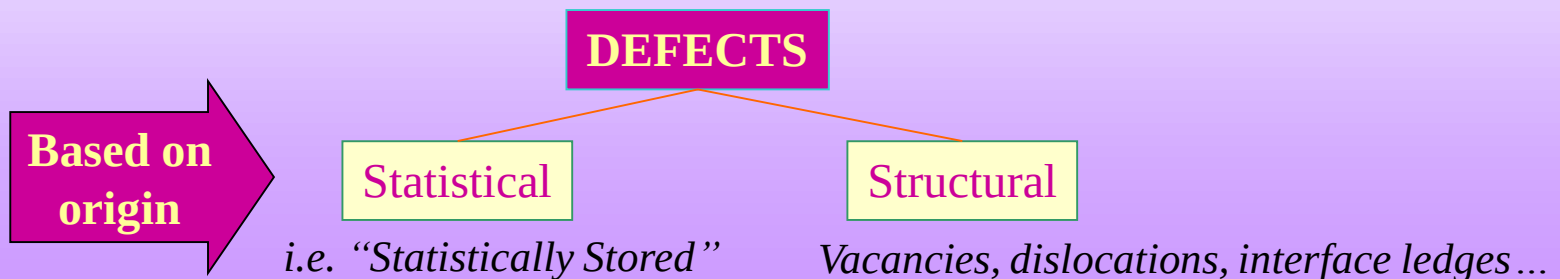
Non-topological

A Defect “*Associated*” with a Symmetry Operation of the Crystal

➤ **TOPOLOGICAL DEFECT**

Statistically stored versus structural defects

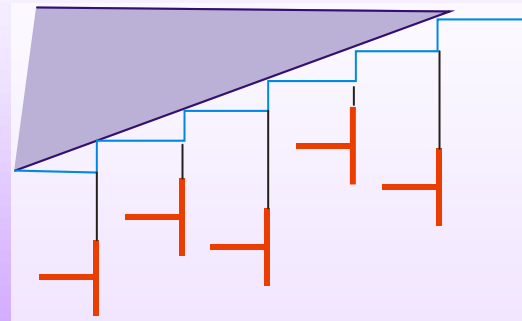
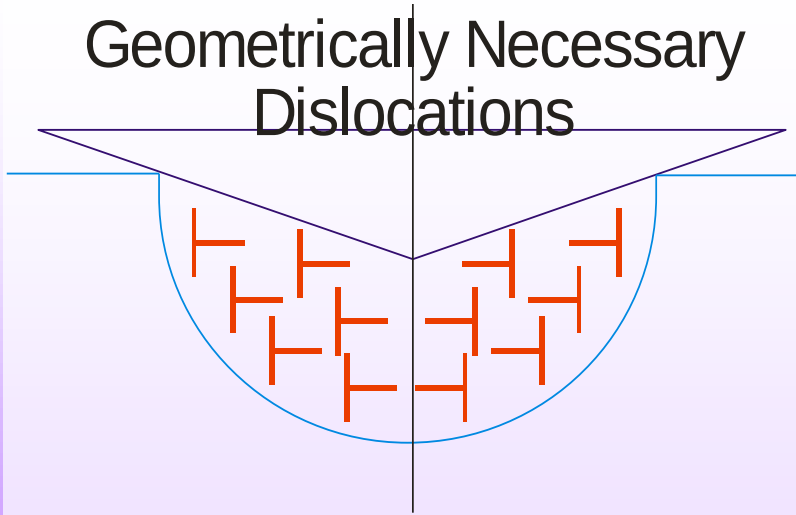
- ❑ A single type of defect (say an edge dislocation) based on its origin may be a **structural defect** (*in which case its location is also determined*) or may be **statistically stored** (*wherein it may be present anywhere in the crystal*).
- ❑ Other equivalent terms to “structural” (*however one should be careful that exact equivalence may not exist in all contexts*) are: “**constitutional**” or “**geometrical necessary**”.
- ❑ **Structural defects** play a very different role in material behaviour as compared to “**Random Statistical Defects**” (*non-structural*).
- ❑ A single defect may play a structural role or may be ‘statistically stored’, based on the context. E.g., a dislocation present within the grain is a statistically stored dislocation, while that present in a low angle grain boundary (LAGB) is a structural dislocation.
- ❑ Structural defects make certain kind of configurations possible in the material (and hence are *localized*). E.g.: angular misorientation between two grains is ‘produced’ by an array of dislocations.
- ❑ A structural dislocation can become a statistically stored one (under some circumstances) and vice-versa is also possible. E.g. a LAGB may absorb a dislocation from the grain and make it a structural dislocation (the GB misorientation may be altered locally).



- ❑ Vacancies (0D). Thermodynamically stabilized vacancies are ‘statistically stored’ (at random positions within the crystal), while those arising from off-stoichiometry are structural.
- ❑ Dislocations (1D). The dislocations arising in the interior of the crystal (say due to faults in crystal growth) are statistically stored, while that at low angle grain boundaries are structural.
- ❑ Terraces/Ledges (2D). Above the roughening transition temperature the surface develops a structure consisting of terraces and ledges— these are statistically stored. Vicinal surfaces have terraces and ledges to accommodate the misorientation with respect to a low index plane— these are structural ledges.

In the example below, geometrically necessary dislocations (GNDs) accommodate the deformation due to indentation.

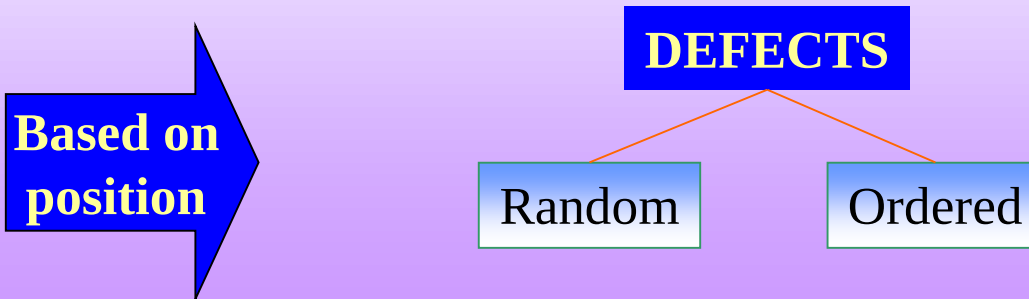
Geometrically Necessary Dislocations



We will see more about these kinds of defects in the relevant chapters.

Random and Ordered Defects

- ❑ In principle any defect can get ordered.
- ❑ Once a defect gets ordered, it needs to be considered part of the structure.
- ❑ The ordering of defects is in principle no different from ordering of other species
 - leads to a **change in symmetry** (and hence can lead to change in crystal structure).
- ❑ Examples include:
 - Vacancy ordering → Vacancy Ordered Phases (VOP)
 - Stacking fault ordering
 - Dislocation ordering.
- ❑ Once ordered, the role of the defect in determining material behaviour will be different.
- ❑ It is important to note that often structural defects are spatially ordered (as well). E.g. dislocations at low angle grain boundaries are structural and they are ordered along the grain boundary.



Q & A

How to understand the difference between the classifications: “**random-ordered**” versus “**statistically-structural**”?

- ❑ In the hyperlink below an example of structural vacancies is considered. They arise due to off-stoichiometry in ordered compounds (say B2 A-B compound: $A_{51}B_{49}$ with vacancies in B-sublattice).
- ❑ Now these vacancies have structural origin, but still are randomly positioned within the B-sublattice.
- ❑ In principle (i.e. not in the example below), these random structural vacancies can get ordered within the B-sublattice; giving rise to a vacancy sublattice. It is to be noted that, this will lower the symmetry of the crystal.
- ❑ An important point to be noted in this context is that, often structural defects (based on origin) are also ordered (based on position). E.g.: (i) dislocations at low-angle grain boundaries are ordered along the grain boundary, (ii) structural ledges on vicinal surfaces are ordered (have an equal spacing), (iii) dislocations at epitaxial interfaces (which are partly coherent), etc.

[Click here to know more about structural/constitutional vacancies](#)

This is the hyperlink

Antisite on Al sublattice ← Ni rich side

NiAl

Al rich side → vacancies in Ni sublattice

Defect in Crystal Structure versus Defect in Property

- ❑ In the chapter on geometry of crystal we have seen that a crystal could be defined based on a geometrical entity (like atoms, molecules) or a physical property (like magnetic moment vector) or both (i.e. the motif could be a geometrical entity, a physical property or both).
- ❑ If the physical property is kept in focus, then the defect could be with respect to the physical property. E.g. in a ferromagnetic material **magnetic moments** are aligned inside the domain and they rotate into a new orientation in a **domain wall** (and hence domain wall is a defect associated with magnetic moment). From a geometrical perspective (atomic positions) the domain wall may have perfect arrangement.

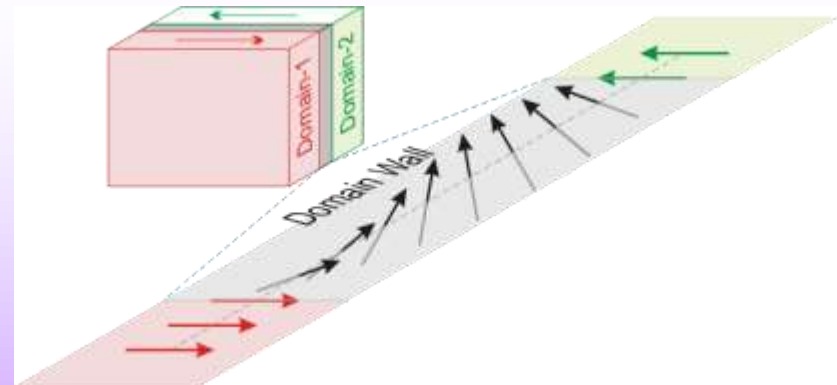
THE ENTITY IN QUESTION

GEOMETRICAL

E.g. atoms, clusters etc.

PHYSICAL

E.g. spin, magnetic moment



Schematic pictures with some defects

Porous Alumina- a 2D crystal

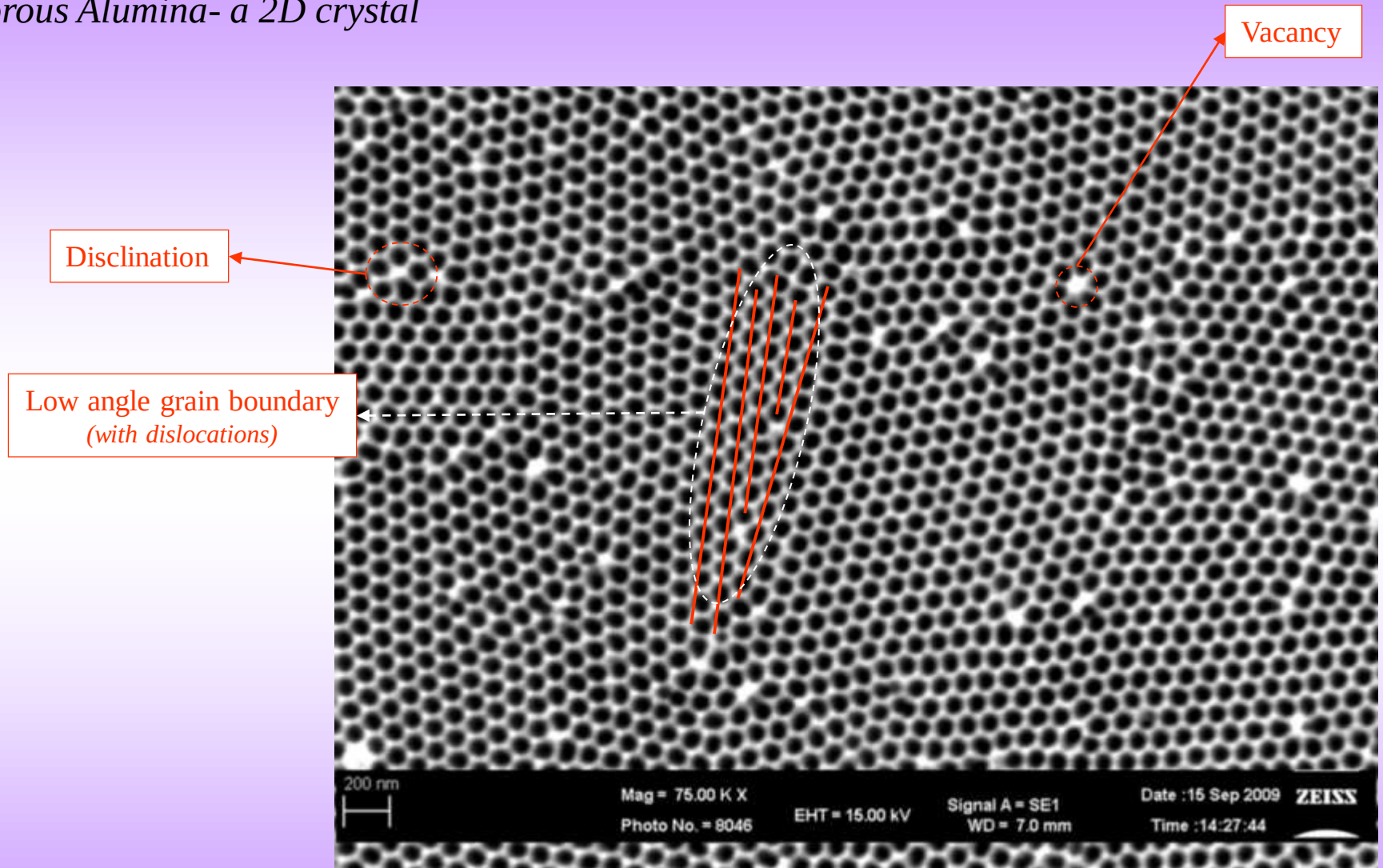


Photo Courtesy- Dr. Sujatha Mahapatra (Unpublished)

Descriptors

- ❑ Often we are not interested in a single defect but, the density of defects. As we have noted before, the dimensionality of these defects vary. The density of these defects will also determine (in a simplistic viewpoint) the average spacing between the defects.
- ❑ ➤ Density of point defects is measured in number (N) per unit volume of the material (V).
- Density of dislocation lines is the total length of dislocation lines (L) per unit volume of the material.
- Density of interfaces (like grain boundaries) is total area of the interface (A) per unit volume of the material.
- Density of 3D objects (like precipitates) is measured as a volume fraction: total volume of objects (V_p) per unit volume of the material.
- ❑ **Important note:** it is a good idea to keep the units as prescribed without canceling the common factors (e.g. the dislocation density should be prescribed in $[m/m^3]$ (and not a $/m^2$) as this preserves the physical meaning).

Dimension	Density		Average spacing (S)	Examples
0	$\rho_0 = \rho_v = N/V$	$[/m^3]$	$S_v \sim (\rho_v)^{-3} [m]$	Vacancy, interstitials
1	$\rho_0 = \rho_d = L/V$	$[m/m^3]$	$S_d \sim (\rho_d)^{-2} [m]$	Dislocation, disclination
2	$\rho_2 = \rho_b = A/V$	$[m^2/m^3]$	$S_b \sim (\rho_b)^{-1} [m]$	Grain boundary, twin boundary
3	$\rho_3 = \rho_p = V_p/V$	$[m^3/m^3]$	$S_\beta \sim (f_\beta)^{1/3} [m]$	Precipitate, dispersoid, void

Key: v-vacancy, d-dislocation, b-boundary, p-particle/void, $(f_\beta)^{1/3}$ - volume fraction

Defects in 2D (Surface) Crystals

- ❑ The diagram below gives an overview of defects in 2D crystals (these are sometimes referred to as surface crystals and should not be confused with surface of crystals).

